

Clementine Zimnicki

+1 224-279-8026 | clemzimnicki@gmail.com | clementinezimnicki.github.io | linkedin.com/in/clementinezimnicki |

PhD candidate studying individual differences in color perception, as well as how the design of information visualizations affects human decision-making under uncertainty, especially in safety-relevant contexts. I have 5+ years of expertise in psychophysical methods, perceptual modeling, and large-scale experimental design.

EDUCATION

University of Wisconsin-Madison

Ph.D. in Psychology

M.A. in Psychology, May 2022

Madison, WI

Aug 2020 – Dec 2026

University of Illinois at Urbana-Champaign

B.S. in Psychology (Cognitive Neuroscience)

Champaign, IL

Aug 2016 – May 2020

TECHNICAL SKILLS

Programming: JavaScript, jsPsych, Python, R, MATLAB, SQL, LaTeX, GitHub

Tools & Software: Adobe CC, GIMP, Microsoft Office Suite, Google Suite, Confluence, Notion, Qualtrics

Communication: quantitative data visualization, scientific writing, teaching & mentoring writing, research talks

Research Methods: human subjects testing, experimental design, survey design, 1:1 interviews, behavioral experiments (2AFC, similarity judgments), psychometric methods, large-scale crowdsourcing (Mechanical Turk, Prolific, CloudResearch), Qualtrics, A/B testing, statistical analysis (mixed-effects models, GLM)

EXPERIENCE

Graduate Research Scientist

University of Wisconsin-Madison

Aug. 2020 – Present

Madison, WI

- Design and execute large-scale psychophysical studies (forced-choice, similarity judgments), collecting 100-1,300+ participants at one time via crowdsourcing platforms
- Perform statistical analyses (power analyses, mixed-effects models, GLM, hierarchical clustering) and generate data visualizations (ggplot, plotly, Adobe Illustrator)
- Present research posters and talks at conferences, communicating findings to cross-disciplinary audiences across engineering, biology, and psychology
- Teach classes of 300+ undergraduate students: lead labs, discussions, and guest lectures; teach data analysis, research methods, and writing; assess individual student progress
- Author peer-reviewed publications, translating complex findings into clear written documents for academic and general audiences

R&D Intern

Sandia National Laboratories

June 2024 – Present

Albuquerque, NM

- Lead studies end-to-end, examining how the colors used in hazard maps affect decision-making under uncertainty
- Design and conduct surveys and interviews to assess apprenticeship program success, analyzing outcomes and translating findings into actionable recommendations for program improvement
- Co-design (with Human Factors, UX, and HR) a workforce skills taxonomy spanning technical and human-factors capabilities to map AI/ML expertise across Sandia and surface gaps in expertise.
- Develop presentations and visualizations distilling research findings into clear, accessible recommendations for higher-level management

Committee Chair

University of Wisconsin-Madison Colloquium Committee

Aug. 2023 – Aug. 2025

Madison, WI

- Directed organization of visits for 4+ invited lecturers per year
- Communicated directly and maintained relationships with donors, budgetary committee, & Psychology department chair

VisXVision Workshop Organizer

Vision Sciences Society & IEEE VIS

Nov. 2022 – Present

Conference locations vary

- Design and lead workshops connecting vision scientists with data vis researchers to facilitate collaboration
- Manage all logistics of event planning, including budget, speaker coordination, event advertising, day-of event execution

HONORS & AWARDS

University Lecture Series Award, **University of Wisconsin-Madison** – 2023

Student Research Grants Competition Funds, **University of Wisconsin-Madison** – 2023

Vision Sciences Society Travel Award, **Vision Sciences Society** – 2023

Kenzi Valentyn Vision Research Grant, **University of Wisconsin-Madison** – 2021

Michael Coles Award for Cognitive Neuroscience Research, **University of Illinois at Urbana-Champaign** – 2019

Carle Neuroscience Institute Undergraduate Research Award, **University of Illinois at Urbana-Champaign** – 2019

PEER-REVIEWED PUBLICATIONS AND CONFERENCE PRESENTATIONS

Zimnicki, C. & Rogers, T.T. Triadic comparisons reveal representational motifs in human color perception. *In prep.*

Zimnicki, C. & Matzen, L. The influence of between- and within-map lightness variation on decision making with hazard maps. *Under Review. Examines how design choices in risk visualizations affect interpretation of hazard severity.*

Zimnicki, C., Tseng, C., Szafr, D. A., & Schloss, K. B. (2023). Effects of data distribution and granularity on color semantics for colormap data visualizations.

Cunningham, E., **Zimnicki, C.**, & Beck, D. M. (2023). The influence of prestimulus 1/f-like versus alpha-band activity on subjective awareness of auditory and visual stimuli.

Zimnicki, C., Szafr, D. A., & Schloss, K. B. (2023). Hue variation masks effects of lightness on interpretations of colormap data visualizations. *Journal of Vision*, 23(9), 5742-5742

Zimnicki, C., & Schloss, K. B. (2022). Squares vs. Maps: effects of configuration on the dark-is-more bias?. *Journal of Vision*, 22(14), 4338-4338.

Zimnicki, C., Cunningham, E., & Beck, D. (2020). Spontaneous pre-stimulus alpha activity predicts visual, but not auditory, detection. *PsychoPhysiology* (Vol. 57, pp. S56-S56).

CONFERENCE WORKSHOPS

Ouxun Jiang, **Clementine Zimnicki**, Kylie Lin, Songwen Hu, & Ghulam Jilani Quadri. (2026). Vis X Vision at Vision Sciences Society 2026.

Ouxun Jiang, **Clementine Zimnicki**, Racquel Fygenon, & Ghulam Jilani Quadri. (2024). Vis X Vision at Vision Sciences Society 2024.

Ghulam Jilani Quadri, **Clementine Zimnicki**, Racquel Fygenon, Madeline Awad, Ouxun Jiang, Cindy Xiong, Danielle Albers Szafr, Christie Nothelfer, & Madison Elliott. (2023). Vis X Vision at IEEE VIS 2023.